

FedCough: A Privacy-Preserving and Resource-Efficient Federated COVID-19 Audio Detection System for Heterogeneous Edge Environments

Design with Differential Privacy, Secure Aggregation,
and Hardware-Aware Optimization

Outline

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COVID-19 and Intelligent Health Monitoring

- ▶ COVID-19 increased demand for distributed, real-time monitoring.
- ▶ Audio (cough, breathing, speech) carries diagnostic signals.
- ▶ Deep learning extracts pathological acoustic biomarkers.

Challenges of Centralized Machine Learning

- ▶ Cough audio is identifiable biometric data (GDPR, HIPAA).
- ▶ Uploading raw audio is risky and bandwidth-heavy.
- ▶ High latency prevents real-time monitoring.

Why Federated Learning?

- ▶ Data stay local; only model updates are exchanged.
- ▶ Reduces privacy risk and communication load.
- ▶ Suitable for diverse medical audio datasets.
- ▶ Challenges: heterogeneity, non-IID data, gradient privacy leakage.

FedCough System Concept

- ▶ End-to-end federated edge intelligence system.
- ▶ Goals: **privacy-preserving**, **efficient**, **robust** COVID-19 detection.

Includes

- ▶ Lightweight audio classifier
- ▶ Differential Privacy
- ▶ Secure Aggregation
- ▶ Hardware-aware adaptation

System Architecture

- ▶ Star-topology federation.
- ▶ Server: scheduling, coordination, aggregation.
- ▶ Clients: local training, DP, compression.

Communication

- ▶ Downlink: quantized global model
- ▶ Uplink: DP + masked updates

Federated Learning Workflow

1. Server selects clients.
2. Broadcast global model.
3. Local training with weighted cross-entropy.
4. Apply DP and secure masking.
5. Upload updates.
6. FedAvg aggregation with sample-size weighting.
7. Hybrid fine-tuning on server.

Concurrency and Fault Tolerance

- ▶ Multi-client simulation using ThreadPoolExecutor.
- ▶ Timeout for slow/failed clients.
- ▶ Semaphore-based resource limit simulation.
- ▶ Prevents straggler-induced stalls.

Differential Privacy Mechanism

- ▶ Gaussian mechanism applied on clients.

Steps

1. Gradient clipping
2. Add Gaussian noise

- ▶ RDP accountant tracks cumulative privacy loss.
- ▶ Clients stop participation when budget is exceeded.

Secure Aggregation via Random Masking

- ▶ Each client adds a high-dimensional random mask.
- ▶ Masked updates look random to the server.
- ▶ Masks cancel out during aggregation.
- ▶ Server only sees aggregated statistics.

Device Profiling

- ▶ Each client assigned:
 - Battery level
 - Compute tier (high/medium/low)
- ▶ Profiles guide training workloads.

Dynamic Training Adaptation

- ▶ Low-resource devices:
 - Reduce local epochs (epoch backoff)
 - Lower computation and tail latency
- ▶ Server adjusts learning rates for compensation.
- ▶ Improves robustness and reduces dropouts.

Quantization and Deployment

- ▶ Post-training INT8 quantization:
 - Min–Max scaling
 - FP32 $\rightarrow$ INT8
 - $\sim 4\times$ size reduction
- ▶ ONNX export for cross-platform deployment.
- ▶ Fidelity check ensures $< 1\%$ accuracy drop.

Experiment 1: Convergence

- ▶ Accuracy: **48.6%** → **65.8%**.
- ▶ Recall: **55.9%** → **66.4%**.
- ▶ Saturation after Round 10.

Experiment 2: Privacy–Utility Trade-off

- ▶ Scenario A (No DP): **65.8%** / **66.4%**.
- ▶ Scenario B ($\epsilon = 5$): **64.6%** / **64.8%**.
- ▶ Scenario C ($\epsilon = 1$): **61.2%** / **62.5%**.
- ▶ Smaller $\epsilon \rightarrow$ more noise $\rightarrow$ lower utility.

Experiment 3: Efficiency and Scalability

Round Time

- ▶ Ours: **18.5s**
- ▶ Baseline: **42.3s**

Dropout Rate

- ▶ Ours: **2.8%**
- ▶ Baseline: **15.8%**

Experiment 4: Compression and Communication

- ▶ FP32: 12.8 MB
- ▶ INT8: **3.3 MB** ($\sim 4\times$ smaller)
- ▶ ONNX: same accuracy, cross-platform compatible

Scalability and Architectural Limitations

- ▶ Star topology becomes a bottleneck at scale.
- ▶ Secure aggregation increases server load.
- ▶ Future: Hierarchical FL (device–edge–cloud).

Differential Privacy Limitations

- ▶ Noise weakens minority-class gradients.
- ▶ Long-tail imbalance worsens DP utility trade-off.
- ▶ Need adaptive privacy budget allocation.

Fairness and Participation Diversity

- ▶ Restricting to high-end devices reduces fairness.
- ▶ Hardware-aware scheduling increases participation.
- ▶ Leads to more robust, generalized models.

Lessons Learned

- ▶ Non-IID data is a major challenge.
- ▶ Device heterogeneity must be addressed.
- ▶ DP creates privacy–utility tension.
- ▶ Simulation $\neq$ real-world behavior.

Future Work

- ▶ Stronger secure aggregation (e.g., FHE).
- ▶ Real-device deployment (Raspberry Pi, Android).
- ▶ Blockchain-based decentralized FL.

Thank You!

Questions?

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